

Aquarium Light DSL

Aquarium Light DSL is a minimal, event-driven language for controlling the lamp of an aquarium. It allows users to declare event handlers using *when ... end*, and to raise events with the syntax *emit event value*. The language works around two events, *outside_light* (the light level measured outside the room) and *hour_changed* (the current hour), enabling behaviours such as dimming the lamp when the room is bright, or using warm light in the evening and cold light during the day.

The language has one simple datatype, number, and one user defined type, the list, for example `[0, 0, 0]`, which stores the latest light measurements so they can be averaged. Values are stored with assignment (`total = total + r`), choices are made with *if... end*, and lists are traversed with *for ... in ... end*.

Input is collected with *read*, and output is done with *show* for messages and with *lamp brightness / lamp color* for the lamp itself, for example *lamp color 20* (warm light, in Celsius).